The rank-four, eleven-element excluded minors for representability over $\mathbb{F}_5$

Computational proof and exact catalogue
31 October 2026

Abstract

An exhaustive computation gives exactly 35 isomorphism classes of rank-four, eleven-element excluded minors for representability over $\mathbb{F}_5$. We give the mathematical justification of the enumeration, its computational evidence, and exact descriptions of all 35 matroids. The main reduction uses compatible representable deletions: on more than twice the rank many elements, the basis-exchange axiom follows from the corresponding axiom in every single-element deletion. Complete sets of labelled representations turn this condition into finite relations on the hyperplanes of a ten-element parent. The computation covers all 58,756 simple representable parent classes, exhausts every required relation problem, and checks all contractions. Each retained class has a separate determinant-based unsatisfiability certificate and explicit representations of its 22 immediate minors. The classification therefore combines a proof of exhaustive coverage with independently checkable certificates for the listed objects.

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1 The classification

All matroids in this paper are finite. A matroid is *quinary* if it is representable over $\mathbb{F}_5 = \mathbb{Z}/5\mathbb{Z}$. An *excluded minor* for quinary matroids is a matroid that is not quinary but every proper minor of which is quinary. Isomorphism always means abstract matroid isomorphism, unless an equivalence of representations is explicitly specified.

Theorem 1.1 (Computational classification)**.** *There are exactly 35 isomorphism classes of rank-four, eleven-element excluded minors for representability over $\mathbb{F}_5$. Representatives $M_1, \ldots, M_{35}$ are specified in Appendix A.*

The proof has two computational parts. To establish that the catalogue contains every required class, we enumerate complete represented parent catalogues and solve an exhaustive finite compatibility problem for each eligible parent. To establish that every retained matroid is an excluded minor, we check a separate certificate of nonrepresentability and representations of every immediate minor. The latter certificates alone would establish only a lower bound of 35 classes; completeness also requires the former computation.

The exact catalogue, in lexicographic basis-bit format, is `results/r4-n11/excluded.jsonl`. Its SHA-256 digest is

`7940989ae051b32496dbfd7e4717d405431bcfd89da69d0dd522517552c76441.`

The identifiers in the appendix refer to this fixed catalogue.

Corollary 1.2. *There are exactly 35 isomorphism classes of rank-seven, eleven-element excluded minors for representability over $\mathbb{F}_5$.*

Proof. Field representability is closed under duality, and duality interchanges deletion and contraction. Thus M is an excluded minor if and only if M^* is. Duality gives a bijection between the rank-four and rank-seven isomorphism classes on eleven elements. □

For context, Brettell's classification through ten elements gives 5, 92, 219, and 130 rank-four excluded minors at sizes seven, eight, nine, and ten, respectively [1]. These four catalogues provide end-to-end calibration instances for the computation here. Compatible extensions and the need to preserve inequivalent representations also appear in the computational methods of Brettell and Pendavingh [2]. Our coverage argument below uses complete represented catalogues and imposes compatibility with every deletion.

2 Preliminaries and structural reductions

For a rank-r matroid M on E, write $\mathcal{B}(M)$ for its set of bases. It is a nonempty family of r-subsets satisfying the ordered basis-exchange axiom: if $A, B \in \mathcal{B}(M)$ and $a \in A \setminus B$, then there is $b \in B \setminus A$ such that $(A \setminus \{a\}) \cup \{b\} \in \mathcal{B}(M)$. We use the usual notation $M \setminus e$ and M/e for deletion and contraction. The *simplification* $\mathrm{si}(M)$ deletes loops and retains one element from each parallel class. A hyperplane of a rank-r matroid is a flat of rank $r-1$.

Lemma 2.1 (Immediate minors suffice)**.** *A matroid M is an excluded minor for quinary matroids if and only if M is not quinary and $M \setminus e$ and M/e are quinary for every $e \in E(M)$.*

Proof. Every proper minor is a minor of an immediate deletion or contraction. Representability over a fixed field is closed under taking minors. The reverse implication follows directly from the definition. □

Lemma 2.2 (Connectivity reduction)**.** *An excluded minor for representability over a field is 3-connected if its ground set has at least four elements. In particular, if M has rank four and eleven elements and is such an excluded minor, then for every $e \in E(M)$ the deletion $M \setminus e$ is a connected, simple, rank-four quinary matroid on ten elements.*

Proof. Representability over a fixed field is closed under direct sums and 2-sums. The standard decomposition of a disconnected matroid as a direct sum, or of a connected matroid with a 2-separation as a 2-sum, has summands isomorphic to proper minors [3]. If all proper minors were representable, these decompositions would make the matroid itself representable, a contradiction. This proves 3-connectivity. For the closure under 2-sums, one may choose representations of the two summands in which their common element is a fixed unit vector, identify that vector in the two vector spaces, and then delete its column. The resulting column matroid represents the 2-sum.

A 3-connected matroid on at least four elements has no loops, coloops, parallel pairs, or series pairs. Hence $N = M \setminus e$ is simple and has rank four. It is quinary because it is a proper minor. To see that it is connected, suppose that $E(N) = A \dot\cup B$ is a nontrivial direct-sum partition. Choose the names so that $|A| \geq 2$. Then $r_M(A) + r_M(B) = r(M)$ and

$$r_M(A) + r_M(B \cup \{e\}) - r(M) \leq 1.$$

Both A and $B \cup \{e\}$ have at least two elements, contradicting 3-connectivity of M. □

The represented parent catalogue includes disconnected as well as connected matroids. Only disconnected parents are discarded before the compatibility search. A deletion of a 3-connected matroid need not be 3-connected, so the reduction requires connected parents, without a 3-connectivity restriction on them.

Lemma 2.3 (Deletion compatibility)**.** *Let E have $n > 2r$ elements and let $\varnothing \neq \mathcal{B} \subseteq \binom{E}{r}$. Suppose that for each $e \in E$ the family*

$$\mathcal{B}_e = \{B \in \mathcal{B} : e \notin B\}$$

is the set of bases of a rank-r matroid on $E \setminus \{e\}$. Then $\mathcal{B}$ is the set of bases of a rank-r matroid on E.

Proof. Take $A, B \in \mathcal{B}$ and $a \in A \setminus B$. Since $|A \cup B| \leq 2r < n$, choose $e \notin A \cup B$. Both A and B are bases of the deletion matroid specified by $\mathcal{B}_e$. Its basis-exchange axiom supplies $b \in B \setminus A$ with $(A \setminus \{a\}) \cup \{b\} \in \mathcal{B}_e \subseteq \mathcal{B}$. This proves every instance of basis exchange; nonemptiness is assumed. □

For the target dimensions, $11 > 2 \cdot 4$. Consequently, a prescribed four-subset family whose eleven deletions are rank-four quinary matroids automatically satisfies the matroid axioms. This observation is what makes the finite table search sufficient. At target sizes seven and eight the inequality fails, and the calibration searches impose the basis-exchange clauses explicitly.

3 Complete represented catalogues

An $r \times k$ matrix over $\mathbb{F}_5$ of full row rank represents the matroid whose bases are the r-subsets of columns with nonzero determinant. A simple rank-r representation is a spanning set of points of $\mathrm{PG}(r-1, 5)$. We represent each projective point by its unique nonzero vector whose first nonzero entry is 1. There are

$$\frac{5^r - 1}{4}$$

such points: 31 in rank three and 156 in rank four.

3.1 Equivalence and generation

Two representations of a fixed labelled matroid are *projectively equivalent* if they differ by invertible row operations and nonzero scaling of individual columns. They are *field-isomorphic* if, in addition, column labels may be permuted. These are the equivalences used by Sage's `is_field_equivalent` and `is_field_isomorphic`, respectively [4]. Since $\mathbb{F}_5$ is a prime field, field automorphisms introduce no further equivalences. Abstractly isomorphic matroids can have inequivalent representations.

Proposition 3.1 (Coverage by represented extensions)**.** *Start with the identity representation I_r. At each size, take every simple rank-preserving represented single-element extension of every retained matrix, and retain one representative of each field-isomorphism class. The resulting level of size k contains every simple rank-r quinary representation on k elements up to field isomorphism.*

Proof. In any such representation, choose a basis and transform its columns to I_r. Order the remaining columns arbitrarily. Appending them in this order gives a chain of simple rank-preserving represented extensions, starting at I_r. Inductively the representation at each stage is field-isomorphic to a retained one. An invertible row map and column scalings carry possible appended projective points bijectively to possible appended projective points; a permutation of old labels extends by fixing the new label. Thus replacing a stage by a retained field-isomorphic representation preserves every possible extension class. This proves the assertion by induction on k. □

The implementation uses Sage's unrestricted `linear_extensions(simple=True)`, followed by exact field isomorphism tests. No cross-ratio restriction is supplied. An abstract basis invariant is used only to select buckets for comparisons; equality of that invariant never substitutes for an isomorphism test. Abstract parent classes are deduplicated separately after the represented level is complete. Keeping only one representation of each abstract matroid during growth would invalidate Proposition 3.1.

3.2 Recovering every labelled representation

The tables in the next section require all projective representations of a fixed labelled matroid N, including representations related by nontrivial automorphisms of N. To obtain them from the unlabelled field classes, align each represented class with the chosen labels of N using one matroid isomorphism, and close the aligned matrices under the full group $\mathrm{Aut}(N)$.

We deduplicate these labelled matrices with the following normal form. Fix an ordered basis $B = (b_1, \ldots, b_r)$ of N, reduce its columns to I_r, and write the remaining block as A. Its support is fixed by the matroid:

$$A_{ij} \neq 0 \quad \Longleftrightarrow \quad (B \setminus \{b_i\}) \cup \{j\} \in \mathcal{B}(N), \qquad j \notin B.$$

Form the bipartite graph with row vertices, nonbasis-column vertices, and one edge for each nonzero entry. Choose a deterministic spanning forest of this graph and scale rows and columns so that all entries on the forest are 1. Scale the basis columns inversely to restore I_r.

Lemma 3.2 (Labelled normal form)**.** *For a fixed ordered basis and a deterministic support forest, the normalized matrix is a complete invariant of labelled projective equivalence. This remains true for disconnected support graphs.*

Proof. Root each tree at a row vertex. Choose its row scale, and successively choose the scale at each new vertex to make the joining entry 1. This normalizes every tree edge. A component consisting of an isolated vertex requires no constraint.

After reducing the same labelled basis to I_r, any projective equivalence is a row scaling together with column scalings: its row map must be diagonal because it preserves every basis-column direction. Between two forest-normalized matrices, the multipliers on each forest edge must have product 1. Along a tree this forces all row multipliers to be a common scalar and all column multipliers its inverse. Hence their product is also 1 on every other edge of that component. All entries agree. Conversely, every operation used in the normalization is a projective equivalence. □

Proposition 3.3 (Complete labelled sets)**.** *One alignment of every unlabelled represented class, followed by closure under generators of* $\mathrm{Aut}(N)$ *and normalization as above, produces every labelled projective representation class of* N*.*

Proof. Every labelled representation belongs, after forgetting its labels, to one of the represented classes. Relative to the chosen alignment of that class, any other alignment differs by an automorphism of N. Closing under group generators therefore reaches it, up to projective equivalence. Lemma 3.2 makes equality testing of the resulting labelled classes exact. □

The automorphism group is computed from a bipartite incidence graph: one vertex color consists of elements, and the other consists of the bases or, when fewer, the nonbases. Incidence gives the edges. A color-preserving graph automorphism induces exactly a matroid automorphism on the element vertices, since the chosen family determines all bases. This justifies using the graph automorphism generators in Proposition 3.3.

4 The finite compatibility problem

Fix a connected simple rank-four quinary matroid N on a ten-element set P. Write e for a new element, and $\mathcal{H}(N)$ for the rank-three hyperplanes of N. For each $H \in \mathcal{H}(N)$ introduce a Boolean coordinate x_H. Its intended value is 1 when the new element lies outside the closure of H in an extension.

For any assignment x, define a four-subset family on $P \cup \{e\}$ by

$$\mathcal{B}(x) = \mathcal{B}(N) \cup \Big\{T \cup \{e\} : T \in \binom{P}{3},\ r_N(T) = 3,\ x_{\mathrm{cl}_N(T)} = 1\Big\}.$$

An arbitrary assignment need not define a matroid. The deletion tables will enforce exactly the compatibility needed for it to do so.

Lemma 4.1 (Hyperplane coordinates are sufficient)**.** *Every rank-preserving matroid extension* M *of* N *by* e *has basis family* $\mathcal{B}(x)$ *for a unique assignment* x*.*

Proof. If $r_N(T) < 3$, then $r_M(T \cup \{e\}) \le 3$, so this four-set is not a basis. If $r_N(T) = 3$ and $H = \mathrm{cl}_N(T)$, then $\mathrm{cl}_M(T) = \mathrm{cl}_M(H)$, because $T \subseteq H \subseteq \mathrm{cl}_M(T)$. Consequently $T \cup \{e\}$ is a basis exactly when $e \notin \mathrm{cl}_M(H)$. This depends only on H. Every rank-three hyperplane contains an independent triple, so every coordinate is determined by the basis family. □

4.1 Represented extension relations

For any fixed labelled simple rank-four quinary matroid Q, let $\mathcal{R}(Q)$ be the finite set of patterns of simple represented rank-preserving extensions of Q by e. Initially a pattern has one bit for each triple T of Q, recording whether $T \cup \{e\}$ is a basis. To construct it, take every labelled representation of Q and append each of the 156 projective points of $\mathrm{PG}(3,5)$ except the directions of columns already used. Compute the determinant bits and merge duplicate patterns.

Proposition 4.2 (Exactness of extension tables)**.** *This construction gives every simple quinary rank-preserving extension of the fixed labelled matroid Q, and no other extension pattern.*

Proof. Every appended point gives a representation; exclusion of used directions ensures simplicity. Conversely, restrict any representation of a simple quinary extension to Q. It is projectively equivalent to one of the complete labelled representatives of Q. Apply that equivalence to the whole representation. Its new column is a nonzero projective point not already used. Its basis pattern therefore occurs in the construction. □

All computations are exact modulo five. In the implementation the determinant on three old columns and an appended vector is evaluated as a linear form in that vector, using the cofactors of the old triple. Dependent triples give zero forms. Independent triples spanning the same hyperplane have proportional forms and hence identical nonzero-status bits. The tables are thus compressed without information loss to hyperplane coordinates.

For each $f \in P$, the matroid $N \setminus f$ has rank four because N is connected. Let

$$C_f = \{\mathrm{cl}_N(T) : T \in \binom{P \setminus \{f\}}{3},\ r_N(T) = 3\} \subseteq \mathcal{H}(N).$$

Relabel the complete representations of $N \setminus f$ to its actual labels, construct $\mathcal{R}(N \setminus f)$, and project its triple bits onto C_f. Call the resulting relation $R_f \subseteq \{0,1\}^{C_f}$. Let $R \subseteq \{0,1\}^{\mathcal{H}(N)}$ be the full relation for N itself. Each projected pattern is checked to be constant on triples with the same closure, and zero on dependent triples. Thus $x|_{C_f} \in R_f$ is precisely the assertion that the proposed deletion avoiding f is a simple rank-four quinary extension of $N \setminus f$.

Define the solution set

$$\mathcal{D}(N) = \left\{x \in \{0,1\}^{\mathcal{H}(N)} : x|_{C_f} \in R_f \text{ for every } f \in P\right\} \setminus R.$$

The representations witnessing the different deletion relations need not be restrictions of a single matrix. Requiring such a matrix would discard precisely the nonrepresentable objects being sought.

Proposition 4.3 (Correctness of the compatibility search)**.** *The assignments in $\mathcal{D}(N)$ correspond bijectively to the simple rank-four nonquinary matroids M on $P \cup \{e\}$ such that $M \setminus e = N$ and every single-element deletion of M is quinary.*

Proof. Take $x \in \mathcal{D}(N)$. The family $\mathcal{B}(x)$ contains $\mathcal{B}(N)$ and is therefore nonempty. Its deletion avoiding e is N. Its deletion avoiding $f \in P$ is the rank-four quinary matroid prescribed by $x|_{C_f} \in R_f$. The deletion compatibility lemma makes $\mathcal{B}(x)$ a matroid basis family. Every pair of elements is contained in one of these simple deletions, so the resulting matroid is simple. Proposition 4.2 shows that it is quinary if and only if $x \in R$, which has been excluded.

Conversely, any matroid with the stated properties has a unique hyperplane assignment by Lemma 4.1. Each quinary deletion places its restricted pattern in R_f, while nonrepresentability puts its full pattern outside R. Thus its assignment lies in $\mathcal{D}(N)$. Uniqueness also proves injectivity. □

Every coordinate occurs in the deletion relations. More precisely, $H \notin C_f$ exactly when f is a coloop of the rank-three restriction $N|H$: removing f then destroys all spanning triples of H. A rank-three matroid has at most three coloops. Hence any four different deletions already cover all coordinates. The program checks full coordinate coverage and imposes all ten relations.

4.2 Exhausting the relations

The principal target search uses an exact natural join of finite relations. Start with the single empty assignment. Given a relation on coordinates S and one on T, replace them by

$$A \bowtie B = \{a \cup b : a \in A,\ b \in B,\ a|_{S\cap T} = b|_{S\cap T}\}.$$

Group rows by their overlap bits and take every compatible pair. By induction, the current relation is exactly the set of assignments satisfying the relations processed so far. Once all coordinates have appeared, rows in R can be removed immediately: later joins only test a full assignment, and cannot change it. Processing all deletion relations therefore gives exactly $\mathcal{D}(N)$.

A separate solver encodes the same relations as shared reduced binary decision diagrams and exact Tseitin clauses. Each node satisfies $v \leftrightarrow (x?h : l)$; requiring the root enforces membership in the finite relation. Full patterns in R are forbidden by blocking clauses. Each satisfying hyperplane assignment is also blocked after enumeration. Final UNSAT then means that this parent's solution set has been exhausted. The two methods implement the same finite relations through different enumeration algorithms.

Time limits and intermediate-size bounds yield the status `UNKNOWN`; they never mean exhaustion. The worker can fall back from a bounded join to SAT. A complete classification requires every eligible parent to end with `EXHAUSTED`. The reported independent SAT/join comparisons concern the calibration searches described in Section 7; the entire target computation is not claimed to have been duplicated with both solvers.

5 Contractions, isomorphism, and completeness

Lemma 5.1 (Testing contractions by simplification)**.** *A matroid is representable over $\mathbb{F}_5$ if and only if its simplification is. Consequently, a rank-four candidate whose deletions are quinary is an excluded minor exactly when the simplification of each contraction is a rank-three quinary matroid.*

Proof. A simplification is a restriction. Conversely, a representation of the simplification extends to the whole matroid by inserting zero columns for loops and nonzero scalar copies of columns for parallel elements. Our candidates are simple, so contraction of any element has rank three. Now apply Lemma 2.1. □

We generate the complete simple rank-three represented catalogues through ten elements using Proposition 3.1. The simplification of a contraction has at most ten elements. At size three it is the free rank-three matroid, represented by I_3; at larger sizes, exact isomorphism comparison with the corresponding rank-three catalogue decides representability. This test

uses a complete catalogue of representable matroids, rather than an assumed list of forbidden rank-three minors.

Candidates passing every contraction test are deduplicated by exact matroid isomorphism. An independent final distinctness check uses canonical labels of the colored element–basis/nonbasis incidence graphs. Two such graphs, with the two vertex types distinguished, are isomorphic exactly when the matroids are isomorphic. The full canonical keys are compared; their SHA-256 digests are merely stored identifiers.

Proposition 5.2 (Global coverage)**.** *Suppose the represented catalogues and their labelled representation sets are complete, every connected ten-element parent's compatibility problem is exhausted, and every contraction is tested as above. Then the isomorphism classes retained by the computation are exactly the rank-four, eleven-element quinary excluded minors.*

Proof. Every retained matroid is nonquinary with quinary deletions by Proposition 4.3, and has quinary contractions by Lemma 5.1. It is therefore an excluded minor.

For the converse, let M be any target excluded minor and choose $e \in E(M)$. By Lemma 2.2, $M \setminus e$ is a connected simple rank-four quinary ten-element matroid. It is isomorphic to a parent N in the complete catalogue. Relabel this deletion to N and label the new element e. By Proposition 4.3, the resulting assignment belongs to $\mathcal{D}(N)$ and hence is enumerated. All its contractions are quinary, so it passes the contraction tests. Isomorphism deduplication retains its class. □

6 Separate certificates for the retained matroids

The preceding argument establishes correctness of the search method. We also certify the excluded-minor property of every retained class using computations separate from the compatibility solver.

6.1 A determinant encoding of representability

Let M be a supplied rank-r matroid, with an ordered basis $B = (b_1, \ldots, b_r)$. Any representation can be put in the form $[I_r \mid A]$ in that column order. Fundamental exchanges determine which entries of A are zero, as in Section 3. Normalize a support forest to 1. Each remaining nonzero entry ranges over $\{1, 2, 3, 4\}$.

For each r-subset S of the ground set let

$$I(S) = \{i : b_i \notin S\}, \qquad J(S) = S \setminus B.$$

These sets have equal size. Expansion on the identity columns shows that the full determinant on S differs only by a sign from $\det A[I(S), J(S)]$, with the empty determinant equal to 1. Impose

$$\det A[I(S), J(S)] \neq 0 \quad \Longleftrightarrow \quad S \in \mathcal{B}(M) \qquad \text{for every } S \in \binom{E(M)}{r}.$$

These conditions are compiled to CNF. A field-valued expression is represented by indicators for its possible values, constrained to have exactly one true indicator. For addition or multiplication, clauses require the output indicator for the exact result modulo five for each possible input pair. Constants are substituted directly. A cached Laplace expansion builds the determinants, and unit clauses enforce their zero or nonzero status. Every maximal minor is constrained, not only selected circuits.

Proposition 6.1 (Correctness of the determinant CNF)**.** *The resulting CNF is satisfiable if and only if M is representable over $\mathbb{F}_5$.*

Proof. A representation can be reduced on B and forest-normalized using invertible row operations and nonzero column scalings. Its entries and all intermediate field operations satisfy the indicator clauses and the required determinant conditions. Conversely, a satisfying assignment gives entries in $\mathbb{F}_5$ and exact values for each arithmetic operation. The identity block has rank r, and the conditions on every maximal minor say that its column matroid has exactly $\mathcal{B}(M)$ as its bases. Hence it represents M. □

For each of the 35 representatives, Glucose produces an UNSAT proof in DRAT format for this CNF. A pinned local build of DRAT-trim [5] checks each proof. The saved artifacts include the exact matroid, the normalization and expression map, the DIMACS formula, the DRAT proof, and the checker's output. A verified DRAT proof establishes unsatisfiability of the supplied formula; Proposition 6.1 supplies its mathematical interpretation.

6.2 Representations of all immediate minors

For each retained M and each e, explicit $\mathbb{F}_5$ matrices represent $M \setminus e$ and M/e. The matrices are recovered from represented catalogues by exact isomorphism, then parallel columns and loops are restored when necessary. Their columns follow the increasing order of the remaining parent labels, renumbered to $0, \ldots, 9$.

Each matrix is checked by two routes: equality of its column matroid with the required minor in Sage, and a separate integer modular Gaussian-elimination implementation. The latter computes the rank and checks the basis status of every subset of the required size. Thus all 22 immediate minors of each representative are checked, giving $35 \cdot 22 = 770$ matrices in total. Together with the checked UNSAT formulas, these witnesses establish the excluded-minor property of each listed class independently of the table-search decisions.

7 Computational evidence

The result recorded here was completed on 27 September 2026. The source code and saved artifacts are part of the accompanying repository. Table 1 gives the complete represented catalogue levels used by the search. The size-four rank-four level is the initial identity representation; all other displayed levels were generated by represented extension.

Size	Rank 3 field	Rank 3 abstract	Rank 4 field	Rank 4 abstract
4	2	2	1	1
5	4	4	3	3
6	11	9	12	11
7	22	18	58	44
8	42	34	460	303
9	92	82	5,001	3,763
10	174	168	64,508	58,756

Table 1: Complete simple represented catalogues. "Field" counts field-isomorphism classes of representations; "abstract" counts matroid isomorphism classes.

At rank four and size ten, alignment and automorphism closure yield 71,547 labelled projective representations of the 58,756 abstract parents. These representations, together with the corresponding nine-element labelled catalogues, supply the extension tables.

The target computation partitions the parent indices into the four half-open ranges in Table 2. Of the 58,756 parents, 84 are disconnected and are excluded by Lemma 2.2. All 58,672 connected-parent problems are exhausted, with no `UNKNOWN` case. The search finds 659 parent-labelled deletion-minimal candidates before the contraction tests and global isomorphism deduplication. This count can include several labellings, or different distinguished deletions, of the same abstract matroid.

Start	Stop	Parents	Candidates	Seconds
0	14,689	14,689	514	1794.54
14,689	29,378	14,689	106	1629.19
29,378	44,067	14,689	29	1430.24
44,067	58,756	14,689	10	1299.00

Table 2: The four exhaustive target-search ranges. Candidate counts are before contraction filtering and global deduplication.

The four search tasks each requested one CPU and 4 GB of memory in the ordinary Slurm partition, with a one-hour outer limit. The longest task took about 30 minutes. The saved workflow identifies the search as array job 884603, the coverage merge as 884612, the nonrepresentation certification as 884613, the minor certification as 884614, and the final audit as 884615. The nonrepresentation and minor certification stages took about 17 and 37 seconds, respectively. These timings describe this run and are not complexity bounds.

The merge checks that the logged parent indices cover the entire catalogue exactly once, that all terminal statuses are justified skips or exhaustion, and that the shards used identical input and critical-source digests. The final audit checks the completed parent levels, certificate coverage and catalogue digest, and pairwise distinctness by incidence-graph canonicalization. It records 35 checked nonrepresentation proofs and 770 checked minor matrices.

7.1 Calibration and cross-checks

At target sizes seven through ten, the same workflow reproduces exactly the published rank-four excluded-minor counts

$\lvert E\rvert$	7	8	9	10
classes	5	92	219	130.

The comparison is by exact matroid isomorphism with the imported published catalogue, and not just by agreement of counts. SAT and the independent finite-table join agree on every connected parent problem for the nine-element target. At the ten-element target their complete sets of 5,991 labelled deletion-minimal candidates agree. Both enumeration methods share the construction of extension tables, so their agreement cross-checks enumeration of the relations; it does not independently reimplement representation-table generation.

7.2 Completion of the classification proof

Proof of Theorem 1.1. The complete represented levels, labelled representation sets, and exhausted parent ranges provide the computational hypotheses of Proposition 5.2. Thus every target excluded-minor class occurs in the retained list. The merge returns 35 classes, and the independent canonical incidence-graph check confirms that they are pairwise nonisomorphic. For each of them the checked determinant CNF proves nonrepresentability, and the 22 checked matrices prove representability of all immediate minors. By Lemma 2.1, each is an excluded minor. The list is therefore complete and has exactly 35 classes. □

None of these 35 matroids is paving: each has a triangle, a circuit of size three. Their basis counts range from 210 to 262. Their triangle-count distribution is

triangles	4	5	6	7	8	10	12	13
classes	3	5	5	10	9	1	1	1

These are properties of the final catalogue, rather than filters assumed during the enumeration.

8 Evidence and reproduction

The mathematical argument depends on exhaustive finite computations and exact arithmetic. Its implementation uses Sage for represented extensions, field and abstract isomorphism, and graph automorphisms; custom code for table generation and joins; PySAT/Glucose for SAT; and DRAT-trim for proof checking. The completed run records Sage 10.9 and, in the separate SAT environment, Python 3.11.16, python-sat 1.9.dev15, and NumPy 2.4.6. DRAT-trim is pinned to upstream commit `2e3b2dc0ecf938addbd779d42877b6ed69d9a985`.

The trusted computational components include represented-parent generation, labelled-orbit construction, table construction, exhaustive relation solving, contraction tests, and exact isomorphism handling. DRAT certificates reduce reliance on the nonrepresentation solver, but do not certify exhaustive parent generation or table construction. The result is a computational proof with replayable evidence, rather than a proof whose entire implementation has been verified in a proof assistant.

The principal evidence files, relative to the repository root, are:

Exact objects. `results/r4-n11/excluded.jsonl` and `results/r4-n11/hyperplanes.json`.

Coverage. `results/r4-n11/summary.json`, together with `runs/r4n11/shard-00/parents.jsonl` through `runs/r4n11/shard-03/parents.jsonl` and their manifests.

Final audit. `results/r4-n11/verification.json` and `results/r4-n11/statistics.json`.

Nonrepresentability. The 35 proof directories under `results/r4-n11/nonrepresentation/`, named `matroid-0000` through `matroid-0034`.

Immediate minors. All 770 matrices in `results/r4-n11/minors/minor-witnesses.jsonl`.

The following scripts implement the successive stages:

1. `scripts/generate_parents.py` generates complete represented rank-four and rank-three levels through size ten.
2. `scripts/prepare_representations.py` produces complete labelled rank-four representation sets at sizes five through ten.
3. `scripts/search_compatible_deletions.py` builds and exhausts the target problems and tests contractions. The required parent ranges partition $[0, 58756)$.
4. `scripts/merge_search.py` checks coverage and deduplicates.

5. `scripts/certify_nonrepresentation.py` and `scripts/certify_minors.py` produce and check the separate certificates.

6. `scripts/final_audit.py` checks the completed result and exports the exact catalogue.

On the original cluster, computation should be submitted through Slurm using the existing `my-sage` and `matroid1` environments. The wrapper `slurm/research.slurm` selects an environment and caps numerical-library threads at one. New runs should use fresh output directories. Full commands and their ordering are given in `docs/REPRODUCING.md`; interpreter paths are specific to the original installation. A completed reproducibility archive is `results/r4-n11-reproducibility.tar.gz`; its adjacent JSON manifest records the archive digest. That archive predates this expository writeup.

For a single class M_i, its proof directory has zero-based index $i-1$. Within an allocation, the checker invocation is

```
vendor/drat-trim/drat-trim FORMULA PROOF
```

where `FORMULA` is that directory's `representation.cnf` and `PROOF` is its `proof.drat`. A successful check prints `s VERIFIED`. The saved minor matrices can be checked directly by `gf5minors.matroid.verify_representation`, without rerunning the search or relying on the represented catalogues.

A Exact catalogue

All representatives have ground set $E = \{0, 1, \ldots, 10\}$ and rank four. For each M_i, let $\mathcal{H}_i$ be the family of subsets listed below. The complete definition is

$$\mathcal{B}(M_i) = \{S \in \binom{E}{4} : S \not\subseteq H \text{ for every } H \in \mathcal{H}_i\}.$$

Thus a four-set is a nonbasis exactly when it is contained in at least one listed set. The listed sets are precisely the hyperplanes of size at least four. Hyperplanes of size three contain no four-set and may be omitted. Every dependent four-set of a rank-four matroid is contained in a hyperplane, which proves that this convention specifies the full basis family. The descriptions have also been checked by reconstructing the basis-bit records from them.

The name M_i abbreviates the stable identifier `GF5-r4-n11-iiii`, with i written in four digits. For example, M_1 is `GF5-r4-n11-0001`; its proof directory is `matroid-0000`. Basis and triangle counts are supplied to make comparison with the machine-readable catalogue easier.

M_1**: 250 bases; 6 triangles.** $\{0,1,2,5,7,9\}$, $\{0,1,4,10\}$, $\{0,2,3,4,8\}$, $\{0,2,6,10\}$, $\{0,3,5,10\}$, $\{0,3,6,9\}$, $\{0,4,5,6\}$, $\{0,6,7,8\}$, $\{0,8,9,10\}$, $\{1,2,3,4,5,6\}$, $\{1,2,5,8\}$, $\{1,2,5,10\}$, $\{1,3,7,9\}$, $\{1,4,7,9\}$, $\{1,6,7,8,9,10\}$, $\{2,3,4,7,10\}$, $\{2,3,4,9\}$, $\{2,6,7,8\}$, $\{2,8,9,10\}$, $\{3,6,7,8\}$, $\{3,8,9,10\}$, $\{4,5,6,7,8\}$, $\{4,5,6,9\}$, $\{4,5,6,10\}$, $\{4,8,9,10\}$, $\{5,8,9,10\}$.

M_2**: 233 bases; 7 triangles.** $\{0,1,2,6\}$, $\{0,1,4,7\}$, $\{0,1,8,9,10\}$, $\{0,2,3,4,5\}$, $\{0,2,7,9\}$, $\{0,3,6,9\}$, $\{0,3,7,10\}$, $\{0,4,6,8\}$, $\{0,5,6,10\}$, $\{0,5,7,8\}$, $\{1,2,3,4,5,6,7\}$, $\{1,2,6,8\}$, $\{1,2,6,9\}$, $\{1,2,6,10\}$, $\{1,4,7,8\}$, $\{1,4,7,9\}$, $\{1,4,7,10\}$, $\{2,3,4,5,8,9,10\}$, $\{6,7,8,9,10\}$.

M_3: **252 bases; 7 triangles.** $\{0,1,2,5,7\}$, $\{0,1,3,6\}$, $\{0,1,4,9,10\}$, $\{0,2,3,4,8\}$, $\{0,2,6,9\}$, $\{0,3,5,7\}$, $\{0,4,5,6,7\}$, $\{0,5,7,8,9\}$, $\{0,5,7,10\}$, $\{0,6,8,10\}$, $\{1,2,3,4,5,6\}$, $\{1,2,5,8\}$, $\{1,2,5,9\}$, $\{1,2,5,10\}$, $\{1,3,6,7\}$, $\{1,3,6,8,10\}$, $\{1,3,6,9\}$, $\{1,7,8,9\}$, $\{2,3,4,7,10\}$, $\{2,3,4,9\}$, $\{2,6,8,10\}$, $\{2,7,8,9\}$, $\{3,5,9,10\}$, $\{3,7,8,9\}$, $\{4,5,6,8,10\}$, $\{4,5,6,9\}$, $\{4,7,8,9\}$, $\{6,7,8,9,10\}$.

M_4: **250 bases; 5 triangles.** $\{0,1,2,5,7\}$, $\{0,1,3,10\}$, $\{0,1,4,8\}$, $\{0,2,3,4,8,9\}$, $\{0,2,6,10\}$, $\{0,3,5,7\}$, $\{0,4,5,7,8\}$, $\{0,4,6,8\}$, $\{0,4,8,10\}$, $\{0,5,6,7,9\}$, $\{0,5,7,10\}$, $\{1,2,3,4,5,6\}$, $\{1,2,5,8\}$, $\{1,2,5,9\}$, $\{1,2,5,10\}$, $\{1,6,7,8,9,10\}$, $\{2,3,4,7,10\}$, $\{2,6,7,9\}$, $\{3,5,8,10\}$, $\{3,6,7,9\}$, $\{4,5,9,10\}$, $\{4,6,7,9\}$.

M_5: **247 bases; 7 triangles.** $\{0,1,2,6,8\}$, $\{0,1,7,9\}$, $\{0,2,3,4,5,7\}$, $\{0,3,6,8,9,10\}$, $\{0,4,6,8\}$, $\{0,5,6,8\}$, $\{0,6,7,8\}$, $\{1,2,3,4,5,6\}$, $\{1,2,6,7\}$, $\{1,2,6,9\}$, $\{1,2,6,10\}$, $\{1,3,7,8\}$, $\{1,4,7,10\}$, $\{1,5,8,9,10\}$, $\{2,3,4,5,8\}$, $\{2,3,4,5,9\}$, $\{2,3,4,5,10\}$, $\{2,7,8,9,10\}$, $\{4,6,7,9\}$, $\{4,8,9,10\}$, $\{5,6,7,10\}$.

M_6: **241 bases; 8 triangles.** $\{0,1,2,7,9,10\}$, $\{0,1,6,8\}$, $\{0,2,3,4,5,8\}$, $\{0,3,7,9,10\}$, $\{0,4,7,9,10\}$, $\{0,5,6,7,9,10\}$, $\{0,7,8,9,10\}$, $\{1,2,3,4,5,6\}$, $\{1,3,8,9\}$, $\{1,4,8,10\}$, $\{1,5,7,8\}$, $\{2,3,4,5,7\}$, $\{2,3,4,5,9\}$, $\{2,3,4,5,10\}$, $\{2,6,8,10\}$, $\{3,6,7,8\}$, $\{4,6,8,9\}$.

M_7: **249 bases; 7 triangles.** $\{0,1,2,6,8\}$, $\{0,1,7,9,10\}$, $\{0,2,3,4,5,7\}$, $\{0,3,6,8,9\}$, $\{0,4,6,8\}$, $\{0,5,6,8,10\}$, $\{0,6,7,8\}$, $\{1,2,3,4,5,6\}$, $\{1,2,6,7\}$, $\{1,2,6,9\}$, $\{1,2,6,10\}$, $\{1,3,7,8\}$, $\{1,4,8,10\}$, $\{1,5,8,9\}$, $\{2,3,4,5,8\}$, $\{2,3,4,5,9\}$, $\{2,3,4,5,10\}$, $\{2,7,8,9,10\}$, $\{3,7,9,10\}$, $\{4,6,7,9,10\}$, $\{5,7,9,10\}$.

M_8: **252 bases; 7 triangles.** $\{0,1,2,5,7\}$, $\{0,1,3,6\}$, $\{0,1,8,9\}$, $\{0,2,3,4,8\}$, $\{0,3,5,7\}$, $\{0,4,5,6,7\}$, $\{0,5,7,8\}$, $\{0,5,7,9,10\}$, $\{0,6,8,10\}$, $\{1,2,3,4,5,6\}$, $\{1,2,5,8,9\}$, $\{1,2,5,10\}$, $\{1,3,6,7\}$, $\{1,3,6,8,9\}$, $\{1,3,6,10\}$, $\{1,4,8,9\}$, $\{1,7,8,9,10\}$, $\{2,3,4,7,9,10\}$, $\{4,5,6,8\}$, $\{4,5,6,9\}$, $\{4,5,6,10\}$, $\{6,7,9,10\}$.

M_9: **255 bases; 6 triangles.** $\{0,1,2,5,7\}$, $\{0,1,3,8\}$, $\{0,1,6,9,10\}$, $\{0,2,3,4,8\}$, $\{0,2,6,9\}$, $\{0,3,5,8\}$, $\{0,3,6,8,9\}$, $\{0,3,7,8,10\}$, $\{0,4,5,6,9\}$, $\{0,6,7,9\}$, $\{1,2,3,4,5,6\}$, $\{1,2,5,8,9\}$, $\{1,2,5,10\}$, $\{1,3,7,9\}$, $\{1,7,8,10\}$, $\{2,3,4,7\}$, $\{2,3,4,9\}$, $\{2,3,4,10\}$, $\{2,6,7,8,10\}$, $\{3,5,9,10\}$, $\{4,5,6,7\}$, $\{4,5,6,8\}$, $\{4,5,6,10\}$, $\{4,7,8,9,10\}$, $\{5,7,8,10\}$.

M_{10}: **239 bases; 8 triangles.** $\{0,1,2,6,8\}$, $\{0,1,4,9,10\}$, $\{0,1,5,7\}$, $\{0,2,3,4,5\}$, $\{0,2,7,9\}$, $\{0,3,6,8,9\}$, $\{0,3,7,10\}$, $\{0,4,6,7,8\}$, $\{0,5,6,8\}$, $\{0,6,8,10\}$, $\{1,2,3,4,5,6,7\}$, $\{1,2,6,9\}$, $\{1,2,6,10\}$, $\{1,5,7,8\}$, $\{1,5,7,9\}$, $\{1,5,7,10\}$, $\{2,3,4,5,8,10\}$, $\{2,3,4,5,9\}$, $\{4,6,7,9\}$, $\{4,6,7,10\}$, $\{5,6,9,10\}$, $\{7,8,9,10\}$.

M_{11}: **212 bases; 10 triangles.** $\{0,1,2,6,8,9\}$, $\{0,1,4,10\}$, $\{0,1,5,7\}$, $\{0,2,3,4,5\}$, $\{0,2,7,10\}$, $\{0,3,6,10\}$, $\{0,3,7,9\}$, $\{0,4,6,7\}$, $\{0,5,8,10\}$, $\{1,2,3,4,5,6,7,8\}$, $\{1,2,6,8,10\}$, $\{1,5,7,9\}$, $\{1,5,7,10\}$, $\{2,3,4,5,9,10\}$, $\{4,6,7,9\}$, $\{4,6,7,10\}$, $\{7,8,9,10\}$.

M_{12}: **242 bases; 8 triangles.** $\{0,1,2,6,8\}$, $\{0,1,3,10\}$, $\{0,1,4,9\}$, $\{0,1,5,7\}$, $\{0,2,3,4,5,9\}$, $\{0,4,6,7,9\}$, $\{0,4,8,9\}$, $\{0,4,9,10\}$, $\{0,5,8,10\}$, $\{1,2,3,4,5,6,7\}$, $\{1,2,6,9,10\}$, $\{1,5,7,8\}$, $\{1,5,7,9\}$, $\{1,5,7,10\}$, $\{2,3,4,5,8\}$, $\{2,3,4,5,10\}$, $\{3,7,8,9,10\}$, $\{4,6,7,8\}$, $\{4,6,7,10\}$.

M_{13}: **254 bases; 5 triangles.** $\{0,1,2,5,7\}$, $\{0,1,4,8\}$, $\{0,2,3,4,8,9\}$, $\{0,3,5,7\}$, $\{0,4,5,7,8\}$, $\{0,4,6,8\}$, $\{0,4,8,10\}$, $\{0,5,6,7,9,10\}$, $\{1,2,3,4,5,6\}$, $\{1,2,5,8\}$, $\{1,2,5,9\}$, $\{1,2,5,10\}$, $\{1,3,7,8\}$, $\{1,6,8,9,10\}$, $\{2,3,4,7,10\}$, $\{2,6,9,10\}$, $\{3,6,9,10\}$, $\{4,6,9,10\}$.

M_{14}: **257 bases; 4 triangles.** $\{0,1,2,5,7,9\}$, $\{0,1,4,10\}$, $\{0,2,3,4,8\}$, $\{0,3,6,7,9,10\}$, $\{0,4,7,9\}$, $\{0,5,6,8\}$, $\{0,7,8,9\}$, $\{1,2,3,4,5,6\}$, $\{1,2,5,8\}$, $\{1,2,5,10\}$, $\{1,3,7,8\}$, $\{1,6,8,9,10\}$, $\{2,3,4,7\}$, $\{2,3,4,9\}$, $\{2,3,4,10\}$, $\{2,6,9,10\}$, $\{4,5,7,8,10\}$, $\{4,6,9,10\}$, $\{5,6,9,10\}$.

M_{15}: **262 bases; 4 triangles.** $\{0,1,2,5,7\}$, $\{0,1,4,10\}$, $\{0,2,3,4,8\}$, $\{0,3,5,7,10\}$, $\{0,4,5,7\}$, $\{0,5,6,7,8,9\}$, $\{1,2,3,4,5,6\}$, $\{1,2,5,8\}$, $\{1,2,5,9\}$, $\{1,2,5,10\}$, $\{1,3,7,9\}$, $\{1,4,7,8\}$, $\{1,6,8,9,10\}$, $\{2,3,4,7\}$, $\{2,3,4,9,10\}$, $\{2,6,8,9\}$, $\{2,7,8,10\}$, $\{3,6,8,9\}$, $\{4,6,7,10\}$, $\{4,6,8,9\}$.

M_{16}: **210 bases; 13 triangles.** $\{0,1,2,7\}$, $\{0,1,4,9,10\}$, $\{0,1,5,8\}$, $\{0,2,3,4,5,6\}$, $\{0,2,8,9\}$, $\{0,3,7,9\}$, $\{0,3,8,10\}$, $\{0,4,7,8\}$, $\{0,6,7,10\}$, $\{1,2,3,4,5,6,7,8\}$, $\{1,2,7,9\}$, $\{1,2,7,10\}$, $\{1,5,8,9\}$, $\{1,5,8,10\}$, $\{2,3,4,5,6,9\}$, $\{2,3,4,5,6,10\}$, $\{4,7,8,9\}$, $\{4,7,8,10\}$, $\{5,7,9,10\}$, $\{6,8,9,10\}$.

M_{17}: **245 bases; 7 triangles.** $\{0,1,2,5,7,9\}$, $\{0,1,4,8\}$, $\{0,2,3,4,8,9\}$, $\{0,2,6,9\}$, $\{0,2,9,10\}$, $\{0,3,5,7\}$, $\{0,4,5,7,8\}$, $\{0,4,6,8\}$, $\{0,4,8,10\}$, $\{0,5,6,7,10\}$, $\{1,2,3,4,5,6\}$, $\{1,2,5,8,10\}$, $\{1,3,7,8,9\}$, $\{1,4,7,9\}$, $\{1,6,7,9\}$, $\{1,7,9,10\}$, $\{2,3,4,7,10\}$, $\{3,5,8,9\}$, $\{3,6,8,9,10\}$.

M_{18}: **242 bases; 6 triangles.** $\{0,1,2,5,8,10\}$, $\{0,1,3,10\}$, $\{0,1,4,9,10\}$, $\{0,1,6,10\}$, $\{0,1,7,10\}$, $\{0,2,3,4,9\}$, $\{0,3,5,7\}$, $\{0,4,5,6,9\}$, $\{0,4,7,9\}$, $\{0,4,8,9\}$, $\{0,6,7,8\}$, $\{1,2,3,4,5,6,7\}$, $\{1,2,5,9\}$, $\{1,7,8,9\}$, $\{2,3,4,8\}$, $\{2,3,4,10\}$, $\{2,7,9,10\}$, $\{3,5,7,8\}$, $\{3,5,7,9\}$, $\{3,5,7,10\}$, $\{3,6,8,9,10\}$, $\{4,5,6,8\}$, $\{4,5,6,10\}$, $\{4,7,8,10\}$.

M_{19}: **244 bases; 8 triangles.** $\{0,1,2,6,7,9\}$, $\{0,1,3,9\}$, $\{0,1,4,8,9\}$, $\{0,1,5,9,10\}$, $\{0,2,3,4,5,7\}$, $\{0,2,7,8\}$, $\{0,2,7,10\}$, $\{0,6,8,10\}$, $\{1,2,3,4,5,6\}$, $\{1,2,6,8\}$, $\{1,2,6,10\}$, $\{1,3,8,10\}$, $\{2,3,4,5,8\}$, $\{2,3,4,5,9\}$, $\{2,3,4,5,10\}$, $\{2,8,9,10\}$, $\{3,6,7,8,9\}$, $\{4,6,7,9,10\}$, $\{5,6,7,9\}$, $\{5,7,8,10\}$.

M_{20}: **256 bases; 6 triangles.** $\{0,1,2,5,7\}$, $\{0,1,3,9,10\}$, $\{0,1,4,6\}$, $\{0,2,3,4,6\}$, $\{0,3,5,7\}$, $\{0,4,5,6,7\}$, $\{0,4,6,8,10\}$, $\{0,4,6,9\}$, $\{0,5,7,8,9\}$, $\{0,5,7,10\}$, $\{1,2,3,4,5\}$, $\{1,2,5,6\}$, $\{1,2,5,8,9\}$, $\{1,2,5,10\}$, $\{1,3,6,7,8\}$, $\{1,4,7,9\}$, $\{1,4,8,10\}$, $\{2,3,4,7\}$, $\{2,3,4,8,10\}$, $\{2,3,4,9\}$, $\{2,6,7,9,10\}$, $\{3,5,6,10\}$, $\{3,5,8,9\}$, $\{4,5,8,9,10\}$, $\{4,7,8,10\}$, $\{5,6,8,9\}$.

M_{21}: **226 bases; 8 triangles.** $\{0,1,2,6,8,9\}$, $\{0,1,4,7\}$, $\{0,1,5,10\}$, $\{0,2,3,4,5\}$, $\{0,2,7,10\}$, $\{0,3,7,9\}$, $\{0,3,8,10\}$, $\{0,4,6,10\}$, $\{0,5,7,8\}$, $\{1,2,3,4,5,6,7\}$, $\{1,2,6,10\}$, $\{1,3,8,10\}$, $\{1,4,7,8\}$, $\{1,4,7,9\}$, $\{1,4,7,10\}$, $\{2,3,4,5,8,9,10\}$, $\{2,7,8,9\}$, $\{3,6,8,10\}$, $\{3,7,8,10\}$, $\{6,7,9,10\}$.

M_{22}: **261 bases; 4 triangles.** $\{0,1,2,5,7\}$, $\{0,1,6,8,9\}$, $\{0,2,3,4,8\}$, $\{0,2,6,9\}$, $\{0,3,5,7\}$, $\{0,3,6,9\}$, $\{0,4,5,7\}$, $\{0,4,6,9\}$, $\{0,5,6,7,9,10\}$, $\{0,5,7,8\}$, $\{1,2,3,4,5,6\}$, $\{1,2,5,8,10\}$, $\{1,2,5,9\}$, $\{1,3,7,10\}$, $\{1,4,7,9\}$, $\{2,3,4,7\}$, $\{2,3,4,9,10\}$, $\{2,7,8,9\}$, $\{3,5,8,9\}$, $\{3,6,7,8\}$, $\{4,7,8,10\}$.

M_{23}: **220 bases; 8 triangles.** $\{0,1,2,6\}$, $\{0,1,4,7\}$, $\{0,1,5,9\}$, $\{0,1,8,10\}$, $\{0,2,3,4,5,10\}$, $\{0,2,8,9\}$, $\{0,3,6,8\}$, $\{0,5,7,8\}$, $\{0,6,7,9,10\}$, $\{1,2,3,4,5,6,7,8\}$, $\{1,2,6,9\}$, $\{1,2,6,10\}$, $\{1,3,9,10\}$, $\{1,4,7,9\}$, $\{1,4,7,10\}$, $\{2,3,4,5,9\}$, $\{3,6,8,9\}$, $\{3,6,8,10\}$, $\{4,8,9,10\}$, $\{5,7,8,9\}$, $\{5,7,8,10\}$.

M_{24}: **220 bases; 8 triangles.** $\{0,1,2,6\}$, $\{0,1,4,7\}$, $\{0,1,8,9,10\}$, $\{0,2,3,4,5\}$, $\{0,2,7,9\}$, $\{0,3,6,9\}$, $\{0,3,7,10\}$, $\{0,4,6,8\}$, $\{0,5,6,10\}$, $\{0,5,7,8\}$, $\{1,2,3,4,5,6,7,8\}$, $\{1,2,6,9\}$, $\{1,2,6,10\}$, $\{1,4,7,9\}$, $\{1,4,7,10\}$, $\{2,3,4,5,9,10\}$, $\{4,6,8,9\}$, $\{4,6,8,10\}$, $\{5,7,8,9\}$, $\{5,7,8,10\}$, $\{6,7,9,10\}$.

M_{25}: **243 bases; 7 triangles.** $\{0,1,2,6\}$, $\{0,1,4,7\}$, $\{0,1,8,9,10\}$, $\{0,2,3,4,5\}$, $\{0,2,7,10\}$, $\{0,2,8,9\}$, $\{0,3,6,10\}$, $\{0,3,8,9\}$, $\{0,4,6,8,9\}$, $\{0,5,7,8,9\}$, $\{1,2,3,4,5,6,7\}$, $\{1,2,6,8\}$, $\{1,2,6,9\}$, $\{1,2,6,10\}$, $\{1,4,7,8\}$, $\{1,4,7,9\}$, $\{1,4,7,10\}$, $\{2,3,4,5,8\}$, $\{2,3,4,5,9,10\}$, $\{3,7,8,10\}$, $\{5,6,8,10\}$, $\{6,7,9,10\}$.

M_{26}: **243 bases; 7 triangles.** $\{0,1,2,6\}$, $\{0,1,4,7\}$, $\{0,1,8,9,10\}$, $\{0,2,3,4,5\}$, $\{0,2,8,9\}$, $\{0,3,6,8,9\}$, $\{0,3,7,10\}$, $\{0,4,6,10\}$, $\{0,4,8,9\}$, $\{0,5,7,8,9\}$, $\{1,2,3,4,5,6,7\}$, $\{1,2,6,8\}$, $\{1,2,6,9\}$, $\{1,2,6,10\}$, $\{1,4,7,8\}$, $\{1,4,7,9\}$, $\{1,4,7,10\}$, $\{2,3,4,5,8\}$, $\{2,3,4,5,9,10\}$, $\{2,7,8,10\}$, $\{5,6,8,10\}$, $\{6,7,9,10\}$.

M_{27}: **228 bases; 12 triangles.** $\{0,1,2,7\}$, $\{0,1,4,9\}$, $\{0,1,5,8,10\}$, $\{0,2,3,4,5,6\}$, $\{0,2,8,9,10\}$, $\{0,3,7,9\}$, $\{0,3,8,10\}$, $\{0,4,7,8,10\}$, $\{0,6,8,10\}$, $\{1,2,3,4,5,6,7\}$, $\{1,2,7,8\}$, $\{1,2,7,9\}$, $\{1,2,7,10\}$, $\{1,3,9,10\}$, $\{1,6,8,9\}$, $\{2,3,4,5,6,8\}$, $\{2,3,4,5,6,9\}$, $\{2,3,4,5,6,10\}$, $\{5,7,8,9\}$, $\{6,7,9,10\}$.

M_{28}: **248 bases; 5 triangles.** $\{0,1,2,5,8\}$, $\{0,1,7,9\}$, $\{0,2,3,4\}$, $\{0,2,7,9\}$, $\{0,3,6,7,9,10\}$, $\{0,4,5,6\}$, $\{0,4,7,8,9\}$, $\{0,5,7,9\}$, $\{1,2,3,4,5,6,7\}$, $\{1,2,5,9,10\}$, $\{1,3,8,10\}$, $\{2,3,4,8\}$, $\{2,3,4,9\}$, $\{2,3,4,10\}$, $\{2,6,8,9\}$, $\{3,6,7,8\}$, $\{4,5,6,8,10\}$, $\{4,5,6,9\}$.

M_{29}: **216 bases; 8 triangles.** $\{0,1,2,9\}$, $\{0,1,6,7,8,10\}$, $\{0,2,3,4,5,10\}$, $\{0,3,8,9\}$, $\{0,4,7,9\}$, $\{0,5,6,9\}$, $\{1,2,3,4,5,6,7,8\}$, $\{1,3,9,10\}$, $\{1,6,7,8,9\}$, $\{2,3,4,5,9\}$, $\{2,7,9,10\}$, $\{4,6,9,10\}$, $\{5,8,9,10\}$.

M_{30}: **256 bases; 5 triangles.** $\{0,1,2,5,7\}$, $\{0,1,4,9,10\}$, $\{0,2,3,4,8\}$, $\{0,2,6,9,10\}$, $\{0,3,5,9,10\}$, $\{0,4,5,6\}$, $\{0,6,7,8\}$, $\{0,7,9,10\}$, $\{0,8,9,10\}$, $\{1,2,3,4,5,6\}$, $\{1,2,5,8,9\}$, $\{1,2,5,10\}$, $\{1,6,7,8,10\}$, $\{2,3,4,7,10\}$, $\{2,3,4,9\}$, $\{2,6,7,8\}$, $\{3,6,7,8,9\}$, $\{4,5,6,7,8\}$, $\{4,5,6,9\}$, $\{4,5,6,10\}$.

M_{31}: **239 bases; 7 triangles.** $\{0,1,2,6,8,9\}$, $\{0,1,4,7\}$, $\{0,2,3,4,5,10\}$, $\{0,3,6,8\}$, $\{0,4,6,8\}$, $\{0,5,6,8\}$, $\{0,5,7,9\}$, $\{0,6,7,8,10\}$, $\{1,2,3,4,5,6,7\}$, $\{1,2,6,10\}$, $\{1,3,9,10\}$, $\{1,4,7,8\}$, $\{1,4,7,9\}$, $\{1,4,7,10\}$, $\{2,3,4,5,8\}$, $\{2,3,4,5,9\}$, $\{2,7,9,10\}$, $\{3,7,8,9\}$, $\{4,6,9,10\}$, $\{5,8,9,10\}$.

M_{32}: **241 bases; 7 triangles.** $\{0,1,2,5,8\}$, $\{0,1,3,7\}$, $\{0,1,4,9,10\}$, $\{0,2,3,4\}$, $\{0,2,6,7,9\}$, $\{0,3,5,8,9\}$, $\{0,3,6,10\}$, $\{0,4,5,6,8\}$, $\{0,5,7,8\}$, $\{0,5,8,10\}$, $\{1,2,3,4,5,6,7\}$, $\{1,2,5,9\}$, $\{1,2,5,10\}$, $\{1,3,7,8,9\}$, $\{1,3,7,10\}$, $\{2,3,4,8,9\}$, $\{2,3,4,10\}$, $\{2,6,7,8,10\}$, $\{3,6,8,9\}$, $\{3,8,9,10\}$, $\{4,5,6,9\}$, $\{4,5,6,10\}$, $\{5,7,9,10\}$.

M_{33}**: 241 bases; 6 triangles.** $\{0,1,2,5,8,9\}$, $\{0,1,3,10\}$, $\{0,2,3,4\}$, $\{0,2,6,10\}$, $\{0,3,6,7,9\}$, $\{0,4,5,6\}$, $\{0,4,7,8\}$, $\{0,4,9,10\}$, $\{0,5,7,10\}$, $\{1,2,3,4,5,6,7\}$, $\{1,2,5,10\}$, $\{1,3,8,9\}$, $\{1,4,8,9,10\}$, $\{1,6,8,9\}$, $\{1,7,8,9\}$, $\{2,3,4,8\}$, $\{2,3,4,9,10\}$, $\{3,6,7,8,10\}$, $\{4,5,6,8\}$, $\{4,5,6,9,10\}$, $\{4,7,9,10\}$.

M_{34}**: 258 bases; 5 triangles.** $\{0,1,2,5,7,9\}$, $\{0,1,4,10\}$, $\{0,2,3,4,8\}$, $\{0,3,5,7\}$, $\{0,3,6,9,10\}$, $\{0,4,5,6,7\}$, $\{0,5,7,8\}$, $\{0,5,7,10\}$, $\{1,2,3,4,5,6\}$, $\{1,2,5,8,10\}$, $\{1,3,7,9\}$, $\{1,4,7,9\}$, $\{1,6,7,8,9\}$, $\{1,7,9,10\}$, $\{2,3,4,7\}$, $\{2,3,4,9\}$, $\{2,3,4,10\}$, $\{2,6,7,10\}$, $\{3,5,8,9\}$, $\{3,7,8,10\}$, $\{4,5,6,8\}$, $\{4,5,6,9\}$, $\{4,5,6,10\}$, $\{4,8,9,10\}$.

M_{35}**: 238 bases; 8 triangles.** $\{0,1,2,6,8\}$, $\{0,1,5,7,9\}$, $\{0,2,3,4,5,10\}$, $\{0,3,6,8,9\}$, $\{0,4,6,7,8\}$, $\{0,5,6,8\}$, $\{0,6,8,10\}$, $\{1,2,3,4,5,6,7\}$, $\{1,2,6,9\}$, $\{1,2,6,10\}$, $\{1,3,9,10\}$, $\{1,5,7,8,10\}$, $\{2,3,4,5,8\}$, $\{2,3,4,5,9\}$, $\{2,7,9,10\}$, $\{4,6,7,9\}$, $\{4,6,7,10\}$, $\{4,8,9,10\}$, $\{5,6,9,10\}$.

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